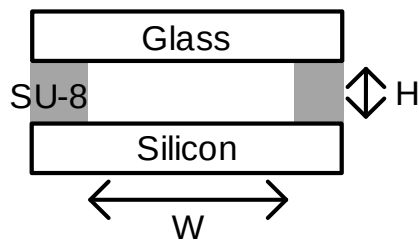


Politecnico di Milano
Biochip A.Y. 2018/2019 - M. Carminati
 Exam of 23.01.2020

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

Consider a microfluidic passive mixer realized in a technology with linear channels fabricated in SU-8 with a thickness H sandwiched between two rigid substrates (silicon and glass). Assume a rectangular cross section, as shown in the figure, with the channel width $W \gg H$.



Data:

$$H = 10 \mu\text{m}$$

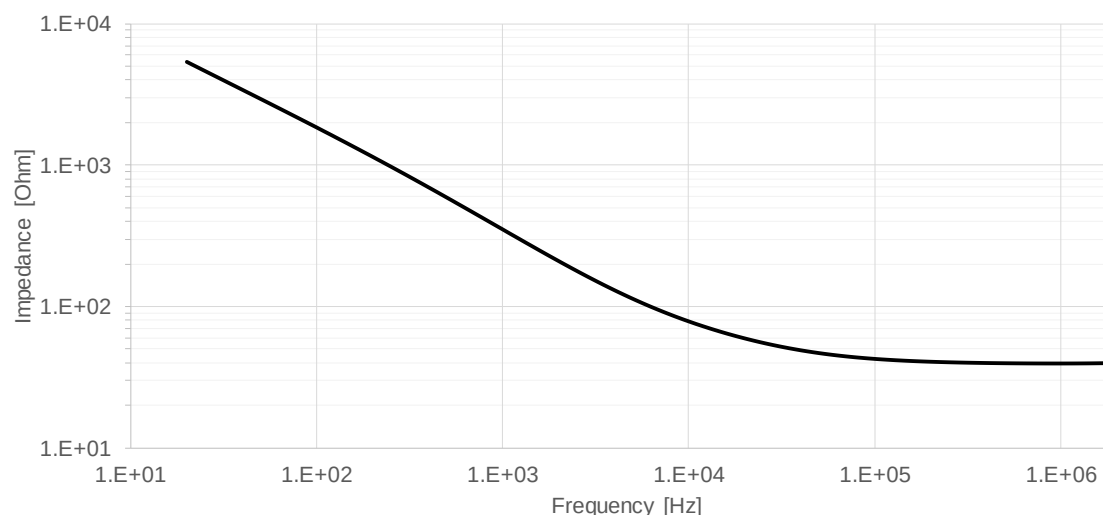
$$1\text{atm} = 101 \text{ kPa}$$

$$\eta_{\text{water}} = 10^{-3} \text{ Pa}\cdot\text{s}$$

- Assuming water-based samples and that the pressure difference applied across the mixer length is 20 Pa, and that the desired speed of fluid in the mixer channel is $1 \mu\text{m/s}$, find the length L and width W of the channel.
- Draw a possible layout of the mixer, offering compactness. How does the geometry change in the case of blood instead of water-based sample?
- Now consider a single straight segment of a channel, in order to achieve the same flow rate Q , is it better (from the point of view of the pressure drop) to realize a single wide channel or to split it into several parallel channels, whose cross-sectional area is the equal to the one of the single wide channel?
- Propose a solution to improve the effectiveness of mixing.
- Describe the fabrication steps of this device.
- Is there a difference in the processing of a glass substrate with respect to a silicon substrate?

Exercise 2

Now consider an interdigitated impedance sensor measuring the growth of bacteria on the surface. The reported spectrum in measured in absence of bacteria, in drinking water with a conductivity of $500 \mu\text{S/cm}$ is reported in the figure.



Data:

$$\epsilon_0 = 8.86 \times 10^{-12} \text{ F/m}$$

$$\epsilon_{r_water} = 80$$

$$\sigma_{\text{PBS}} = 1.5 \text{ S/m}$$

$$C_{\text{DL}} = 0.1 \text{ pF}/\mu\text{m}^2$$

- Draw the impedance equivalent model of the interface and estimate the values of the model parameters from the plot.
- Redraw a quoted spectrum in the case conductivity is increased to $750 \mu\text{S/cm}$.
- Redraw a qualitative spectrum in the case bacteria deposit on the sensor surface.
- Choose the optimal measuring frequency to electrically track the growth of the bacteria colony.
- In order to optimize sensitivity to a single layer of bacteria, what should be the optimal gap between electrodes?
- Draw a scheme of the electronic circuit for measuring the impedance at the chosen frequency. Choose all the values (voltages, passive components, etc...) of the designed scheme.